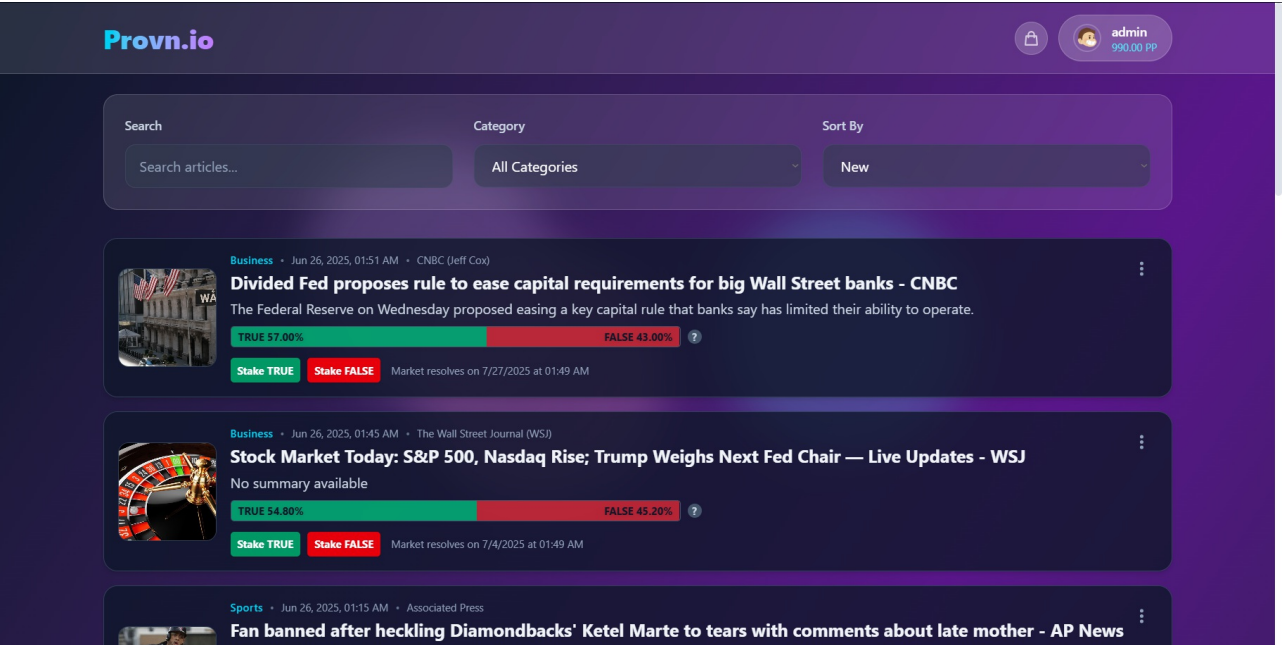


Provn.io: Crowdsourcing news verification



Provn.io is a prediction markets platform for news verification that harnesses the collective efforts of online users to dramatically scale verification efforts through prediction markets. We mobilise users by giving them a stake in truthfulness, and use rewards and incentives to ensure the accuracy of predictions made by these simulated markets.

All your news, in one place



Deep dive into each article, while tracking how much readers trust its claims

Provn.io

admin990.00 PP

Brandy and Monica reunite for first-ever joint tour, more than 25 years after hit duet "The Boy Is Mine" - CBS News

CBS News • By Analisa Novak • June 24, 2025 at 08:18 PM



Grammy-winning R&B stars Brandy and Monica announced their 24-city arena tour on "CBS Mornings."

What do you think about this story's truthfulness?

Current Odds

TRUE50.0%

FALSE50.0%

Market Stats

Total Stakes:0

Volume:0.00 PP

Created:2 days ago

Place Your Stake

Your Prediction

TRUEFALSE

Stake Amount (Available: 990.00 PP)

10

Stake 10.00 PP on TRUE

<div>
<h2>Help verify articles for others while you browse, and get rewarded!</h2>
</div>

Provn.io

admin990.00 PP

Search

Search articles...

Category

New

Sort By

New



Business • Jun 26, 2025, 01:51 AM • CNBC

Divided Fed proposes rule

The Federal Reserve on Wednesday

TRUE 57.00%

Stake TRUE Stake FALSE

Market resolves on 7/4/2025 at 01:49 AM



Business • Jun 26, 2025, 01:45 AM • The V

Stock Market Today: S&P

No summary available

TRUE 54.80%

Stake TRUE Stake FALSE

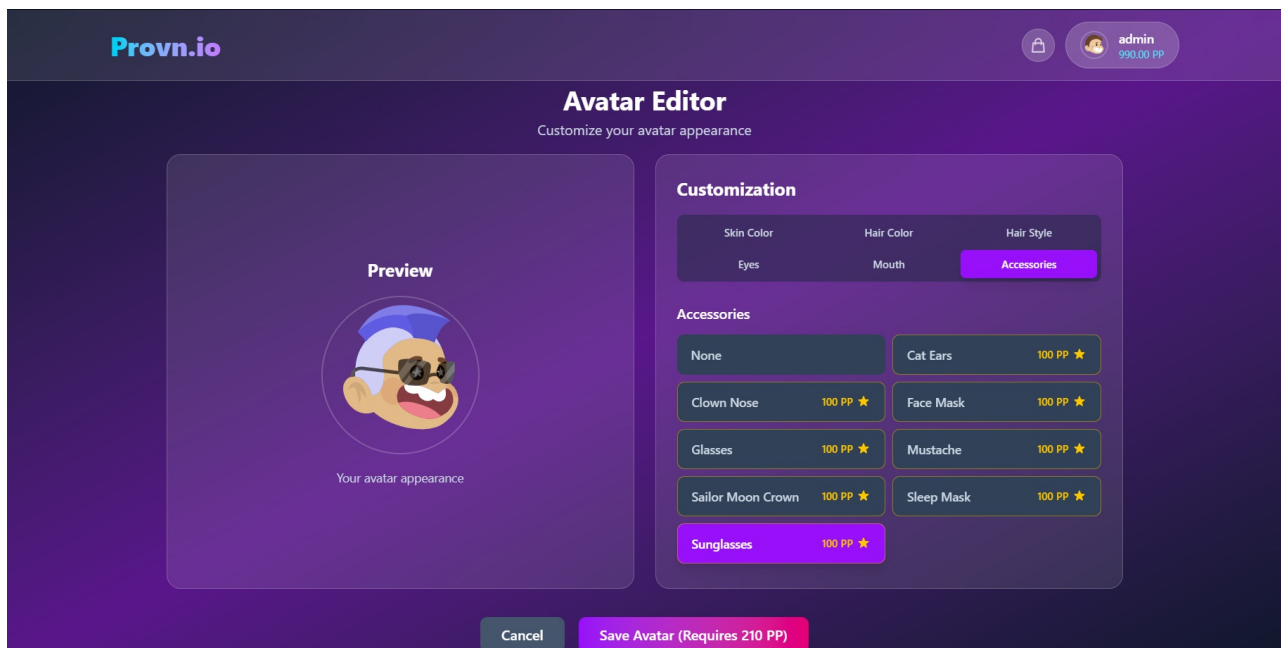
Market resolves on 7/4/2025 at 01:49 AM



Sports • Jun 26, 2025, 01:15 AM • Associated Press

Fan banned after heckling Diamondbacks' Ketel Marte to tears with comments about late mother - AP News

<div>
<h2>Accumulate ProvePoints with your predictions to unlock new profiles!</h2>
</div>



🔗 Core Features

👤 User Accounts & Authentication

- **User Account Creation:** Robust signup with email validation, password strength requirements, and duplicate prevention
- **Secure Authentication:** JWT-based authentication with proper password hashing, input validation, and protection against common attacks
- **Profile Management:** User profile pages with personalized statistics and staking history
- **Admin Features:** Administrative controls for determining veracity of new articles and resolving markets as needed

📰 News Integration & Article Management

- **Real-time News Fetching:** Integration with NewsAPI.org for fetching real-world news articles
- **Article Display:** Rich article cards with source attribution, publication dates, and category tags
- **Diverse Categories:** Articles across business, entertainment, health, science, sports, and technology
- **Search & Sorting:** Filter articles by keyword search and sort for new, trending, trusted, or contentious articles

📊 Prediction Markets System

- **Automatic Market Creation:** Prediction markets are automatically created for each news article, allowing users to predict TRUE or FALSE on news article claims
- **Completely Liquid Markets:** Users place stakes with the Provn.io platform directly, instead of staking against other users, which allows stakes to be placed quickly and easily with Provn.io's automated market making
- **Real-time Probability Updates with LMSR:** Market probabilities predicting the veracity of news with the Logarithmic Market Scoring Rule update dynamically based on user stakes
- **Market Statistics:** Comprehensive statistics showing total participants, stake amounts, and probability distributions
- **Market Resolution:** Admin-controlled market resolution with automatic payout calculation

💰 ProvePoints Economy

- **Virtual Currency:** ProvePoints (PP) serve as the platform's staking currency
- **Starting Balance:** New users receive 100 PP to begin participating
- **Stake-based Wagering:** Users stake PP on their predictions with dynamic upside calculations
- **Winning Payouts:** Automatic distribution of winnings based on market outcomes and LMSR calculations
- **Balance Management:** Real-time balance updates and insufficient funds protection

👤 Avatar Customization System

- **DiceBear Integration:** Avatar system powered by DiceBear's "big-smile" style
- **Comprehensive Customization:** Skin color, hair color, hair style, eyes, mouth, and accessories
- **Real-time Preview:** Instant preview of avatar changes in the editor
- **Global Display:** Avatars appear throughout the app (navbar, profile, stakes history)
- **Premium Features:** PP-gated premium avatar options, rewarding users for correct predictions with cosmetic features to customise their profile

🔧 Tech Stack

Frontend

- **React 19** with TypeScript for type-safe component development
- **Tailwind CSS** for modern, responsive styling with custom glass-morphism effects
- **Vite** for fast development builds and hot module replacement
- **React Router** for client-side routing and navigation
- **React Hot Toast** for user notifications and feedback

Backend

- **Node.js** with Express.js for RESTful API development

- **TypeScript** for full-stack type safety
- **Prisma ORM** with SQLite database for data persistence
- **JWT** for stateless authentication
- **bcrypt** for secure password hashing
- **CORS** for cross-origin resource sharing

External Integrations

- **NewsAPI.org** for real-time news article fetching
- **DiceBear API** for avatar generation and customization

Getting Started

Prerequisites

- Node.js (version 16 or higher)
- npm or yarn package manager
- SQLite3 (for database)

Installation & Setup

1. Clone the repository

```
git clone <repository-url>
cd provn-orbital25
```

2. Install dependencies

```
npm run install:all
```

This installs dependencies for the root, backend, and frontend.

3. Environment Configuration

Backend Setup:

- Copy backend/.env.example to backend/.env
- Configure your environment variables:

```
DATABASE_URL="file:./prisma/dev.db"
JWT_SECRET="your-jwt-secret-key"
NEWS_API_KEY="your-newsapi-key-from-newsapi.org"
```

News API Setup:

- Visit [NewsAPI.org \(https://newsapi.org/\)](https://newsapi.org) and create a free account
- Get your API key and add it to the .env file
- Free tier provides 1,000 requests per day

4. Database Setup

```
cd backend
npx prisma generate
npx prisma db push
```

5. Create Initial Data

```
# Create admin user (username: admin, password: password123)
node scripts/createAdminUser.js

# Create test user (username: testuser, password: password123)
node scripts/createTestUser.js

# Fetch real news articles and create markets
node scripts/fetchRealNews.js
```

6. Start Development Servers

```
# From root directory - starts both backend and frontend
npm run dev

# Or start individually:
npm run dev:backend # Backend on http://localhost:3000
npm run dev:frontend # Frontend on http://localhost:5173
```

7. Access the Application

- **Frontend:** <http://localhost:5173>
- **Backend API:** <http://localhost:3000>
- **Database Browser:** Run `npx prisma studio` in backend directory (<http://localhost:5555>)

Available Scripts

Root Level:

- `npm run dev` - Start both backend and frontend in development mode
- `npm run build` - Build both backend and frontend for production
- `npm run test` - Run tests for both backend and frontend
- `npm run install:all` - Install dependencies for all packages

Backend Scripts:

- `npm run dev` - Start backend development server with hot reload
- `npm run build` - Build backend for production
- `npm test` - Run backend tests

Frontend Scripts:

- `npm run dev` - Start frontend development server
- `npm run build` - Build frontend for production
- `npm run preview` - Preview production build

User Accounts & Testing

Test User Accounts

For testing and development purposes, you can use these pre-configured accounts:

Regular Test User

- **Username:** testuser
- **Email:** test@example.com
- **Password:** password123
- **Starting Balance:** 100 PP

Admin User

- **Username:** admin
- **Email:** admin@example.com
- **Password:** password123
- **Starting Balance:** 10,000 PP
- **Admin Features:** Market resolution, user management

These accounts come pre-configured with initial ProvePoints and can be used to:

- Test the authentication system
- Explore prediction markets and staking
- Test avatar customization features
- View performance analytics
- (Admin only) Resolve markets and manage content

Creating Your Own Account

You can also register a new account through the registration page, which provides:

- 100 PP starting balance
- Full access to all platform features
- Persistent data storage
- Avatar customization options

Avatar System

Provn.io features a comprehensive avatar customization system powered by DiceBear's "big-smile" style:

Avatar Features

- **Customizable Options:** Skin color, hair color, hair style, eyes, mouth, and accessories
- **Premium Features:** Unlock premium options with ProvePoints (50 PP for hair styles, 30 PP for eyes/mouth, 100 PP for accessories)
- **Real-time Preview:** See changes instantly in the avatar editor
- **Persistent Storage:** Avatar configurations are saved to your profile
- **Global Display:** Avatars appear throughout the app (navbar, profile, stakes history)

How to Customize Your Avatar

1. Navigate to your Profile page or click "Avatar Shop" in the navigation
2. Click on your current avatar image or use the Avatar Editor page
3. Choose from different customization categories (skin, hair, eyes, mouth, accessories)
4. Premium features require sufficient ProvePoints to unlock
5. Save your changes to update your profile across the platform

Understanding Prediction Markets

How Markets Work

1. **Article Integration:** Every news article automatically gets a prediction market
2. **Binary Outcomes:** Predict whether a news claim will prove TRUE or FALSE
3. **Dynamic Pricing:** Market probabilities update in real-time based on user stakes
4. **ProvePoints Staking:** Use PP to place stakes on your predictions
5. **Market Resolution:** Admins resolve markets based on real-world outcomes
6. **Automatic Payouts:** Winners receive payouts calculated using LMSR formulas

Staking Process

1. **Choose a Market:** Browse articles and select a market to participate in
2. **Make Your Prediction:** Choose TRUE or FALSE for the article's claim
3. **Set Stake Amount:** Decide how many ProvePoints to stake

4. **View Potential Returns:** See your potential upside multiplier and winnings
5. **Confirm Stake:** Submit your prediction to the market
6. **Track Performance:** Monitor your stakes in your profile page

⚙️ API Endpoints

Authentication Endpoints

- POST /api/auth/register - Register new user account
- POST /api/auth/login - User login with email/username and password
- GET /api/auth/profile - Get current user profile (protected)
- PUT /api/auth/profile - Update user profile (protected)
- PUT /api/auth/update-avatar - Update user avatar configuration (protected)

Article Endpoints

- GET /api/articles - Get all articles with optional filtering
- GET /api/articles/categories - Get available article categories
- POST /api/articles/refresh - Fetch new articles from News API (admin)

Market Endpoints

- GET /api/markets - Get all prediction markets
- GET /api/markets/:id - Get specific market by ID
- GET /api/markets/:id/staking-parameters - Calculate staking parameters for a prediction
- PUT /api/markets/:id/resolve - Resolve market outcome (admin)
- PUT /api/markets/:id/admin-resolve - Admin resolve market (admin)

Stake Endpoints

- POST /api/stakes - Create new stake (protected)
- GET /api/stakes/user - Get current user's stakes (protected)
- GET /api/stakes/market/:marketId - Get stakes for specific market
- GET /api/stakes/stats/:marketId - Get market statistics

User Endpoints

- GET /api/users/me - Get current user details (protected)
- GET /api/users/stats - Get platform statistics

🗄️ Database Schema

The application uses SQLite with Prisma ORM. Key models include:

User Model

- Authentication data (username, email, password)
- ProvePoints balance and admin status
- Avatar configuration (skin, hair, eyes, mouth, accessories)
- Timestamps and reset tokens

Article Model

- News article content (title, description, URL, image)
- Source attribution and publication date
- Category classification
- Associated prediction market

Market Model

- Article association and outcome tracking
- LMSR parameters (shares, probabilities)
- Resolution status and timestamps
- Associated stakes

Stake Model

- User prediction and stake amount
- Upside calculation and resolution status
- Market and user associations
- Creation timestamp

🔑 Development & Testing

Testing the System

1. Authentication Testing

```
cd backend
node test-auth.js
```

2. News API Testing

```
cd backend
node scripts/testNewsAPI.js
```

3. Stake Integration Testing

```
cd backend
node scripts/testStakeIntegration.js
```

Database Management

- **View Database:** Run `npx prisma studio` in **backend** directory
- **Reset Database:** Delete `backend/prisma/dev.db` and run `npx prisma db push`
- **Backup Database:** Copy `backend/prisma/dev.db` file

Environment Variables

Ensure your `backend/.env` file contains:

```
DATABASE_URL="file:./prisma/dev.db"
JWT_SECRET="your-secure-jwt-secret-key"
NEWS_API_KEY="your-newsapi-key"
PORT=3000
```

 Project Structure

```
provn-orbital25/
├── backend/                # Node.js Express API server
│   ├── prisma/            # Database schema and migrations
│   ├── scripts/          # Utility scripts for data management
│   └── src/
│       ├── controllers/   # API route handlers
│       ├── middleware/    # Authentication and validation
│       ├── routes/        # API route definitions
│       ├── services/      # Business logic (Markets, Stakes, Users)
│       ├── tests/         # Unit and integration tests
│       └── public/        # Static files and test pages
├── frontend/              # React TypeScript application
│   ├── public/            # Static assets
│   └── src/
│       ├── components/    # Reusable UI components
│       ├── contexts/      # React context providers
│       ├── pages/         # Page components
│       ├── services/      # API communication
│       ├── utils/         # Helper functions and types
│       └── shared/        # Shared TypeScript types
```

 Production Deployment

Environment Setup

1. Set production environment variables
2. Configure production database (PostgreSQL recommended)
3. Set up proper JWT secrets and security headers
4. Configure News API for production limits

Build Process

```
npm run build
```

Security Considerations

- JWT secret should be cryptographically secure
- Database should be properly backed up
- Rate limiting should be implemented for API endpoints
- Input validation is enforced on all user inputs

 Future Enhancements

Planned Features

- **Multi-timeframe Markets:** 1-day, 1-month, and 5-month prediction windows
- **Community Forums:** Discussion spaces for each article
- **Advanced Analytics:** Detailed performance metrics and leaderboards
- **Social Features:** Follow users, share predictions, and collaborative analysis
- **Mobile App:** Native mobile applications for iOS and Android
- **Enhanced Avatar System:** More customization options and unlockable content

Technical Improvements

- Real-time updates using WebSockets
- Enhanced caching for better performance
- Advanced search and recommendation algorithms
- Integration with additional news sources
- Automated market resolution using AI/ML

 Contributing

We welcome contributions to Provn.io! Here's how you can help:

1. **Fork the repository**

2. **Create a feature branch** (`git checkout -b feature/amazing-feature`)
3. **Commit your changes** (`git commit -m 'Add some amazing feature'`)
4. **Push to the branch** (`git push origin feature/amazing-feature`)
5. **Open a Pull Request**

Development Guidelines

- Follow TypeScript best practices
- Maintain test coverage for new features
- Use consistent code formatting (Prettier/ESLint)
- Update documentation for API changes
- Test thoroughly before submitting PRs

LMSR Logic in Our Prediction Markets

Our platform uses the **Logarithmic Market Scoring Rule (LMSR)** as the automated market maker for all prediction markets. LMSR is a popular mechanism for prediction markets because it provides continuous liquidity and automatically adjusts prices (probabilities) based on the stakes placed by users. This ensures that users can easily place stakes at any time, while also allowing the platform to detect shifts in sentiment—**serving as an early warning system for potential misinformation**.

How LMSR Works

- **Shares System:**
In a traditional LMSR market, users buy and sell "shares" in possible outcomes. The price of each share is determined by the current distribution of shares and the market's liquidity parameter. As more shares are bought for an outcome, its implied probability (and price) increases.
- **Price Calculation:**
The LMSR formula ensures that the cost to move the market probability increases as the market becomes more certain. This prevents any single user from drastically moving the market with a small stake.

User-Friendly Abstraction

To make the experience more intuitive, we **abstract away the concept of shares**. Instead, users interact with the market using **ProvePoints (PP)**, our platform's staking currency.

- **Direct Upside Visualization:**
When a user stakes ProvePoints on an outcome, the platform calculates and displays their **expected upside** (potential gain per PP staked) using the underlying LMSR math. This means users don't need to understand or manage shares—they simply see how much they stand to gain if their prediction is correct.
- **Behind the Scenes:**
 - When a user stakes, the system uses LMSR to determine how many "virtual shares" their stake would buy.
 - The market's probabilities (odds) are updated based on the new share distribution.
 - The user's upside is calculated and shown directly, making the process transparent and user-friendly.

Example

Suppose the market is 50/50 and you stake 100 PP on "Yes":

- The system calculates, via LMSR, how much this moves the probability.
- You immediately see your **upside per PP** and the new market odds.
- No need to worry about shares or complex formulas—just stake and see your potential reward.

License

MIT (<https://choosealicense.com/licenses/mit/>)

Authors and Acknowledgments

- Joel Tan Zhuo Yao - Full-stack development, system architecture, and market mechanics
- Yap Yulun - Frontend development, UI/UX design, and user experience

Provn.io - Harnessing collective intelligence for news verification through prediction markets.